METRIC
MIL-PRF-85045/22A
w/AMENDMENT 1
13 August 2018
SUPERSEDING
MIL-PRF-85045/22A
17 June 2014

PERFORMANCE SPECIFICATION SHEET

CABLE, FIBER OPTIC, EIGHTEEN FIBERS, STANDARD AND ENHANCED PERFORMANCE, CABLE CONFIGURATION TYPE 2 (OFCC), APPLICATION B (SHIPBOARD), CABLE CLASS SM AND MM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-85045.

CLASSIFICATION:

Fiber optic cable configuration type: 2 (OFCC)

Fiber optic cable class: MM (graded-index, glass core and glass cladding, multimode)

SM (dispersion-unshifted, glass core and glass cladding, single-mode)

DESIGN AND CONSTRUCTION:

Fiber: Class MM fibers shall be in accordance with MIL-PRF-49291/6.

Class SM fibers shall be in accordance with MIL-PRF-49291/7.

Buffer diameter: 900±50 µm

OFCC: Dimensions and configuration: See figure 1.

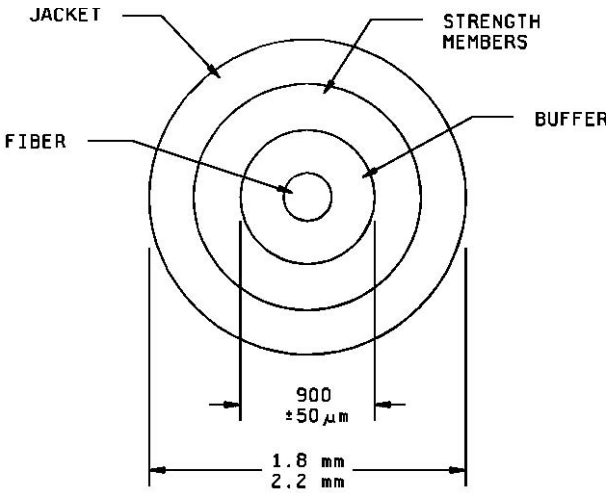


FIGURE 1. Optical fiber cable component.

Mass per unit length: <15 kg/km

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Tensile loading: >270 N

Dynamic bend tensile load: 90 N minimum

Jacket material: The OFCC jacket shall be composed of a low halogen, low smoke, low toxicity polymer material.

Cable jacket color: Shall be black or blue.

OFCC color:

Slate (MIL-PRF-49291/6 fiber).

Yellow (MIL-PRF-49291/7 fiber).

OFCC marking: Each OFCC shall be uniquely marked with a number between 1 and 18. The form of the marking shall be the printed spelling of the number, followed by a dash, followed by the printed Arabic numeral. The marking shall be applied and repeated every 0.10 m along the OFCC jacket.

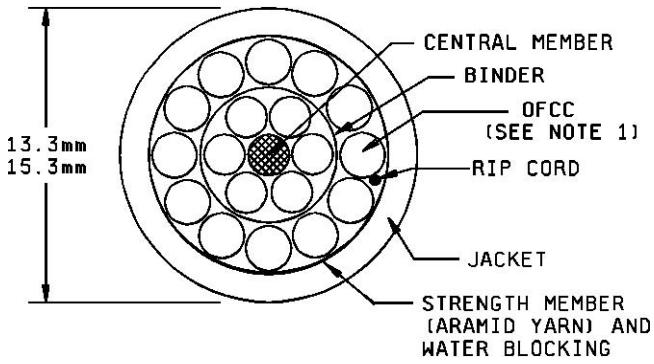
OFCC marking for mixed fiber cables: OFCCs containing MIL-PRF-49291/6 shall be marked between number 1 and the number representing the total number of MIL-PRF-4921/6 fibers in the cable. OFCCs containing MIL-PRF-49291/7 shall be marked from the lowest number available, after all OFCCs containing MIL-PRF-49291/6 have been marked, up to 18.

Short-term minimum bend diameter: Eight times the OFCC outer diameter.

Long-term minimum bend diameter: Sixteen times the OFCC outer diameter.

Finished cable:

Dimensions and configuration: See figure 2. Eighteen OFCC units shall be helically laid over the central member. OFCC number one shall be located in the innermost ring of OFCCs. OFCCs shall be consecutively numbered starting with those in the innermost ring (1 through 6) and then those in the outer ring (7 through 18). OFCCs 1 and 7 shall be located along approximately the same radial line within the cable. The minimum outer jacket thickness shall be not less than 1.4 mm.



NOTE:

1. OFCC – Optical fiber cable component

FIGURE 2. Eighteen OFCC fiber optic cable.

Number of fibers: 18 (one per OFCC)

Concentricity: >0.65

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Mass per unit length: <200 kg/km

Jacket material: The overall jacket shall be composed of a low halogen, low smoke, low toxicity polymer material.

Short-term minimum bend diameter: Eight times the cable outer diameter. (The short-term minimum bend diameter is to be used in all environmental and mechanical tests which specify a cable minimum bend diameter.)

Long-term minimum bend diameter: Sixteen times the cable outer diameter.

Minimum continuous length: The minimum continuous length of all cables shall be not less than 0.5 km. If lengths less than 0.5 km are specified in the purchase order, a conformance inspection shall be performed on test specimens not less than 0.5 km in length from which the purchase order lengths are cut.

PERFORMANCE REQUIREMENTS:

Optical properties:

Maximum attenuation rate (Standard Performance):	4.5 dB/km at 850±20 nm, 2.0 dB/km at 1300±20 nm for class MM fiber. 1.5 dB/km at 1310±20 nm and 1550±20 nm for class SM fiber.
Maximum attenuation rate (Enhanced Performance):	5.0 dB/km at 850±20 nm, 2.0 dB/km at 1300±20 nm for class MM fiber. 1.5 dB/km at 1310±20 nm and 1550±20 nm for class SM fiber.

For cables with radiation cross-linked jackets, the change in attenuation rate measurement may be made up to 30 days after cross-linking of the cable jacket.

Bandwidth: Fiber with a minimum bandwidth of 500 MHz-km at 1300 nm and 160 MHz-km at 850 nm shall be used (multimode cables only).

Change in optical transmittance: Measurements shall be made at 1300±20 nm. A minimum of eight fibers shall be monitored for change in optical transmittance. A minimum of two fibers shall be monitored in the inner ring of OFCCs. A minimum of six fibers shall be monitored in the outer ring of OFCCs. The fibers monitored in each ring shall be randomly selected, but different fibers shall be monitored in each test sample. For shock testing, only four fibers are required to be monitored (two in the inner ring and two in the outer ring of OFCCs).

Mechanical properties:

Tensile loading and elongation: Applicable, tensile loading ≥3300 N.

Low temperature flexibility: Applicable. For enhanced cable jackets the exposure temperature shall be -40 °C.

Cyclic flexing: 500 cycles at 25±2 °C and 100 cycles at -28±2 °C. Change in optical transmittance measurements are to be made every 100 cycles for the 500-cycle exposure and every 25 cycles for the 100-cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Cable twist-bending: 500 cycles at 25±2 °C and 100 cycles at -28±2 °C. Change in optical transmittance measurements are to be made every 100 cycles for the 500-cycle exposure and every 25 cycles for the 100-cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Impact: 50 cycles at 25±2 °C and 20 cycles at -40±2 °C.

Cable jacket tear strength: Applicable, except the cable jacket tear strength shall be not less than 35 N/cm of jacket thickness for standard performance cable.

Hosing: Both low pressure and hydrostatic pressure are applicable.

Low pressure: Leakage shall be not greater than 60 ml.

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Hydrostatic: 2.1 MPa

Cable abrasion resistance:

Standard performance product: Applicable.

Enhanced performance product:

Cable scraping resistance: 750 cycles

Cable-to-cable abrasion: 500 cycles

Environmental properties:

Temperature range:

Operating: -28 to 65 °C

Nonoperating: -40 to 70 °C

Storage: -40 to 70 °C

Temperature cycling: Change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made over a period of 1 hour at the end of each temperature plateau.

Temperature humidity cycling: Change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made at the end of each temperature plateau.

Temperature life (life aging): Applicable.

Standard performance product: The test shall be conducted as specified in MIL-PRF-85045.

Enhanced performance product: The test shall be conducted as specified in MIL-PRF-85045 except that the jacket material shall be tested at +175 °C for 4 hours.

Fluid immersion: Applicable.

Standard performance product: Exposure to automobile gasoline and tap water are not required and the test temperature for lubricating oil exposure shall be 73 to 77 °C. The tensile strength retention of the cable jacket material after exposure to hydraulic fluid shall be not less than 30 percent.

Enhanced performance product: Exposure to automobile gasoline and tap water is not required and the following test temperatures shall be used for the fluids indicated: fuel oil (98 to 100 °C), turbine fuel (48 to 50 °C), and lubricating oil (98 to 100 °C).

Chemical properties:

Halogen content: <0.2 percent

Cross-link verification: This test is applicable for cables with cross-linked jackets only. The test shall be conducted in accordance with ICEA standard T-28-562 and run at 200 °C. The test shall be in the qualification test sequence using a specimen cut from the sample used in the thermal shock test and in the Group III testing of the Group C conformance test sequence. The hot creep elongation shall not exceed 100 percent, and the hot creep set shall not exceed 10 percent.

Identification marking: For mixed fiber cables, the identification marking shall include the following: 18 Fiber - A by 62.5/125, B by 9/125, where A=number of 62.5/125 fibers and B=number of 9/125 fibers.

Part or identifying number (PIN):

Standard performance

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M85045/22-01 (Multimode)

M85045/22-02 (Single-mode)

Enhanced performance

M85045/22-01E (Multimode)

M85045/22-02E (Single-mode)

M85045/22-03E-XX XX (Mixed Fiber: XX Multimode fibers and XX Single-mode fibers)

- XX: 49291/6 fiber count
- XX: 49291/7 fiber count

Qualification and conformance inspection: See table I.

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TABLE I. Qualification and conformance inspection.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length 1/ 2/	Conformance inspection
I	Visual and mechanical inspection	3.4, 3.9, 3.10	4.7.2	3 samples, 0.5 km each 3/	A
	Attenuation rate	3.5.1	4.7.4.1	3 samples, 0.5 km each 4/	A
II	Crosstalk	3.5.3	4.7.4.3	3 samples, 0.5 km each 4/	--
	Electromagnetic effects	3.7.16	4.7.6.16	1 specimen, 6 m 4/	--
III	Temperature cycling	3.7.1	5/	2 samples, 0.5 km each 4/ (1 on reel, 1 off)	C
	Temperature humidity cycling	3.7.3	4.7.6.3	2 samples, 0.5 km each 6/	C
	Storage temperature	3.7.4	4.7.6.4	2 samples, 0.5 km each 6/	--
	Low temperature flexibility (cold bend)	3.6.3	4.7.5.3	2 specimens, 8 m each 7/	B
	Cyclic flexing	3.6.4	4.7.5.4	6 specimens, 5 m each 8/ (2 specimens at each temp)	--
	Crush	3.6.5	4.7.5.5	2 specimens, 5 m each 7/	--
	Cable twist-bend	3.6.6	4.7.5.6	6 specimens, 5 m each 8/ (3 specimens at each temp)	C
	Impact	3.6.8	4.7.5.8	2 specimens, 5 m each 6/	C
	Temperature life (life aging)	5/	5/	2 specimens, 300 m each 7/	C
	Tensile loading and elongation	3.6.1	4.7.5.1	2 specimens, 150 m each 9/	C
	Operating tensile loading	3.6.1.1	4.7.5.1.1	2 specimens, 150 m each 10/	C
	Fungus resistance	3.8.4	4.8.4	2 specimens, 0.5 m each 7/	--
	Cable element removability	3.6.18	4.7.5.18	2 specimens, 0.5 m each 7/	C
IV	Thermal shock	3.7.2	4.7.6.2	1 specimen, 0.49 km each 4/ (on reel)	--
	Weathering	3.7.8	4.7.6.8	1 specimen, 2 m 11/ and 3 material samples 12/	--

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TABLE I. Qualification and conformance inspection – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length [1/] [2/]	Conformance inspection
IV	Fluid immersion	3.7.9	4.7.6.9	1 specimen, 2 m [11/] and 3 material samples [12/] for each specified fluid	C
	Paint susceptibility	3.7.15	4.7.6.15	2 specimens, 2 m [13/]	--
	Jacket self-adhesion or blocking	3.7.11	4.7.6.11	1 specimen, 30 m [11/]	--
	Shock	3.7.13	4.7.6.13	1 specimens, 30 m [11/]	--
	Dynamic bend	3.6.2	4.7.5.2	1 specimen, 150 m [14/]	B
	Hosing: low pressure	3.6.12.1	4.7.5.12.1	1 specimen, 1.5 m [11/]	C
	Hosing: hydrostatic	3.6.12.2	4.7.5.12.2	1 specimen, 2 m [11/]	C
	Radial compression (for application B only)	3.6.7	4.7.5.7	1 specimen, 10 m [11/]	--
	Corner bend	3.6.9	4.7.5.9	2 specimens, 5 m [13/]	--
	OFCC kink	[5/]	[5/]	2 specimens, 1 m [13/]	--
	Cross-link verification	[5/]	[5/]	2 specimens, 1 m [13/]	C
V	Dripping	3.6.13	4.7.5.13	1 specimen, 30 cm [11/]	--
	Cable jacket tear strength	3.6.14	4.7.5.14	5 flat extruded jacket material strips [15/]	C
	Cable jacket material tensile strength and elongation	3.6.15	4.7.5.15	5 specimens [16/]	--
	Cable abrasion resistance	3.6.16	4.7.5.16	4 specimens, 2 m [17/]	C
	Cable shrinkage	3.6.17	4.7.5.17	3 specimens, 0.5 m [13/]	C
	Durability of identification	3.6.19	4.7.5.19	3 specimens, 2 m each [13/]	--
	Flame extinguishing and smoke generation	3.7.12.2	4.7.6.12.2	1 specimen, 50 m [11/]	C
	Water absorption	3.7.14	4.7.6.14	2 specimens, extruded jacket material strips [16/]	--
VI	Acid gas generation	3.8.1	4.8.1	1 specimen, 1 m [18/]	C
	Halogen content	3.8.2	4.8.2	1 specimen, 1m [18/]	--
	Toxicity index	3.8.3	4.8.3	1 specimen, 1 m [18/]	C

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TABLE I. Qualification and conformance inspection – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length [1/] [2/]	Conformance inspection
VI	Smoke index	3.8.5	4.8.6	1 specimen, 1 m [18/]	C

NOTES:

1/ Tolerance on 0.5-km length is ±5 percent, provided that results are normalized to 1 km.
2/ Tolerance on shorter lengths is ±5 percent.
3/ The visual and mechanical inspection shall only be conducted on a 2-m section of each sample.
4/ The same samples as used in the visual and mechanical inspection shall be used.
5/ As stated under the applicable tests in this specification sheet.
6/ The same samples as used in the temperature cycling test shall be used.
7/ A specimen cut from each sample used in the temperature cycling test shall be used.
8/ Three specimens cut from each sample used in the temperature cycling test shall be used.
9/ A specimen cut from each specimen used in the temperature life test shall be used.
10/ The same specimen as used in the tensile loading and elongation test shall be used.
11/ A specimen cut from the specimen used in the thermal shock test shall be used.
12/ Three flat extruded jacket material strips as specified in 4.7.5.15 shall be used in this test.
13/ Specimens cut from the specimen used in the thermal shock test shall be used.
14/ A specimen cut from the specimen used in the thermal shock test or one of the specimens used in the tensile loading and elongation test shall be used.
15/ Flat extruded jacket material strips (i.e., strips of flat extruded material with the same composition of the cable jacket and of sufficient dimensions in which dumbbells can be cut) prepared as specified in 4.7.5.14 cited method and obtained from flat extruded material that underwent the thermal shock test shall be used.
16/ Flat extruded jacket material as specified in 4.7.5.15 shall be used in this test.
17/ Specimen cut from the specimen used in the thermal shock test shall be used. Two specimens shall be used for scraping abrasion, and two specimens shall be used in the cable-to-cable abrasion testing.
18/ A specimen cut from one of the samples used in the temperature cycling test or specimen used in the thermal shock test shall be used.

Manufacturers who are qualified for M85045/17-01 or M85045/18-01 and MIL-PRF-85045/20-01, and whose cable passes the visual and mechanical, and attenuation rate inspections specified herein, are qualified under this specification sheet for enhanced performance cable (M85045/22-01E). This qualification by similarity is applicable only if the same OFCCs and same cable materials used in the previously qualified cables are used in the cable under test. Testing may be performed on a single length of cable, with a minimum length of 0.5 km.

Manufacturers who are qualified under M85045/17-01 or M85045/18-01, and whose cable passes the tests and inspections specified in table II, are qualified under this specification sheet for enhanced performance cable (M85045/22-01E). This qualification by similarity is applicable only if the same OFCCs and same cable materials used in the previously qualified cable are used in the cable under test. Testing may be performed on a single length of cable, with a minimum length of 0.5 km.

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TABLE II. Qualification and conformance by similarity to M85045/17-01 or M85045/18-01.

Group	Qualification inspection	Qualification inspection 1/	Conformance inspection 2/
I	Visual and mechanical	X	A
	Attenuation rate	X	A
III	Low temperature flexibility (cold bend)	X	B
	Impact (low temperature only)	X	C
IV	Hosing: low pressure	X	C
	Cable abrasion resistance	X	C
V	Flame extinguishing and smoke generation	X	C

NOTES:

1/ Where manufacturers are qualified for M85045/17-01 or M85045/18-01 and are pursuing qualification for M85045/22-01E.

2/ Where manufacturers are qualified for M85045/17-01 and M85045/22-01 and are performing conformance testing for M85045/22-01E.

Manufacturers who are qualified under this specification sheet for multimode (M85045/22-01E) fiber cable and whose single-mode (multimode) fiber cable passes the tests and inspections specified in table III are qualified under this specification sheet for single-mode (M85045/22-02E) fiber cable. This qualification by similarity is applicable if the only difference between the previously qualified cable and the cable under test is that the optical fiber has been changed from a multimode fiber to a single-mode fiber. Testing may be performed on either one or two lengths of cable, each with a minimum length of 0.5 km. Test order must be observed up to and including the storage temperature test. If only one cable length is used, the thermal shock test shall be performed after the storage temperature test.

Manufacturers who are qualified under this specification sheet for multimode fiber cable (M85045/22-01E) and single-mode fiber cable (M85045/22-02E) and whose mixed fiber cable passes the visual and mechanical, and attenuation rate inspections specified herein, are qualified under this specification for mixed fiber cable (M85045/22-03E). This qualification by similarity is applicable only if the same OFCCs and same cable materials used in the previously qualified cables are used in the cable under test. Testing may be performed on a single length of cable, with a minimum length of 0.5 km.

TABLE III. Qualification and conformance by similarity (fiber).

Group	Inspection	Qualification inspection M85045/22-02E 1/	Conformance inspection M85045/22-02E 2/	Qualification inspection M85045/22-03E 3/	Conformance inspection M85045/22-03E 4/
I	Visual and mechanical	X	A	X	A
	Attenuation rate	X	A	X	A
III	Temperature cycling	X	C	--	--
	Temperature humidity cycling	X	C	--	--
	Storage temperature	X	--	--	--
	Cyclic flexing	X	--	--	--
	Crush	X	--	--	--
	Cable twist-bending	X	C	--	--

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TABLE III. Qualification and conformance by similarity (fiber) – Continued.

Group	Inspection	Qualification inspection M85045/22-02E 1/	Conformance inspection M85045/22-02E 2/	Qualification inspection M85045/22-03E 3/	Conformance inspection M85045/22-03E 4/
III	Impact (low temperature only)	X	C	--	--
	Tensile loading and elongation	X	C	--	--
	Operating tensile loading	X	C	--	--
IV	Thermal shock	X	--	--	--
	Dynamic bend	X	B	--	--

NOTES:
1/ Where manufacturers are qualified for MIL-PRF-85045/22-01E and are pursuing qualification for MIL-PRF-85045/22-02E.
2/ Where manufacturers are qualified for MIL-PRF-85045/22-01E and MIL-PRF-85045/22-02E and are performing conformance testing for M85045/22-02E.
3/ Where manufacturers are qualified for MIL-PRF-85045/22-01E and MIL-PRF-85045/22-02E and are performing conformance testing for M85045/22-03E.
4/ Where manufacturers are qualified for MIL-PRF-85045/22-01E, MIL-PRF-85045/22-02E, and MIL-PRF-85045/22-03E and are performing conformance testing for M85045/22-03E.

AMENDMENT NOTATIONS: The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
NASA – NA

Preparing activity:
Navy – SH
(Project 6015-2018-001)

Review activities:
Army – AR, AV, MI
Navy – EC, YD
Air Force – 02, 19, 99
DLA – CC

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